

# Md Hasebul Hasan

Ph.D. Student in Computer Science and Engineering | Trustworthy, Human-Centered, and Efficient AI  
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## Research Interests

Human-AI interaction, natural language processing, trustworthy AI, LLM evaluation and safety, agentic systems, and efficient LLM inference. I develop methods and benchmarks that diagnose model failures, improve reliability in real-world interactions, and reduce the computational cost of deployment.

## Education

**Ph.D. in Computer Science and Engineering** **Aug 2025 – Present**  
*University of Texas at Arlington, Arlington, TX* *GPA: 4.00/4.00*  
Advisor: Mohammad Atiqul Islam. Graduate Research Assistant.

**M.Sc. in Computer Science and Engineering** **Jul 2022 – Aug 2025**  
*Bangladesh University of Engineering and Technology (BUET), Dhaka* *GPA: 3.75/4.00*  
Thesis: *An Efficient Compositional Reasoning Approach for Solving Geospatial Queries Using LLMs*. Research led to publications at ICML, EACL, and IEEE COMPSAC.

**B.Sc. in Computer Science and Engineering** **Feb 2017 – May 2022**  
*Bangladesh University of Engineering and Technology (BUET), Dhaka* *GPA: 3.77/4.00; Rank: 31/127*  
Senior-year GPA: 3.89/4.00. Thesis: *A Multi-Modal Deep Learning Based Approach for House Price Prediction*.

## Research Experience

**Graduate Research Assistant** **Aug 2025 – Present**  
*University of Texas at Arlington* *Advisor: Mohammad Atiqul Islam*

- Research trustworthy, human-centered, and efficient LLM systems, with emphasis on real-world evaluation, small-language-model adaptation, and memory-efficient inference.
- Created **DeEscalWild**, a benchmark derived from real-world police-civilian interactions containing **1,500 scenarios, 285,887 dialogue turns, and approximately 4.7M tokens**; designed the transcription, diarization, filtering, anonymization, validation, and evaluation pipeline.
- Designed controlled evaluations across multiple 2B–3B language models, including fine-tuning, automatic metrics, LLM-as-a-judge evaluation, expert human review, and latency measurement.
- Investigate KV-cache compression for Llama-2-7B on NVIDIA H100 GPUs, studying low-rank projection, token retention, oracle allocation, K/V asymmetry, memory, throughput, and latency trade-offs.

**Graduate Researcher – GeoAI and LLM Agents** **2022 – 2025**  
*Bangladesh University of Engineering and Technology* *M.Sc. research*

- Co-developed **MapEval**, a benchmark for geospatial reasoning in foundation models; led literature-gap analysis and helped design the spatial reasoning taxonomy, dataset construction, annotation, and human verification.
- Led **MapAgent**, a hierarchical agent with dynamic map-tool integration for compositional geospatial reasoning; first author on the EACL 2026 Findings paper.
- Developed **GuardFL**, a majority-consensus approach for detecting and removing malicious clients in federated learning; first author on the IEEE COMPSAC 2024 paper.

## Publications

### Peer-Reviewed / Accepted

1. M. L. Dihan, M. T. Hassan, M. T. Parvez, **M. H. Hasan**, M. A. Alam, M. A. Cheema, M. E. Ali, and M. R. Parvez. “MapEval: A Map-Based Evaluation of Geo-Spatial Reasoning in Foundation Models.” *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. **Spotlight (top 2.6%)**. [paper]
2. **M. H. Hasan**, M. L. Dihan, T. Hashem, M. E. Ali, and M. R. Parvez. “MapAgent: A Hierarchical Agent for Geospatial Reasoning with Dynamic Map Tool Integration.” *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2026. [preprint]
3. **M. H. Hasan**, M. S. I. Khan, M. S. Haque, A. Y. Ahmed, and M. A. Adnan. “GuardFL: Detection and Removal of Malicious Clients Through Majority Consensus.” *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 1112–1117, 2024. [paper]

## Manuscripts / Preprints

4. **M. H. Hasan**, K. H. Charu, E. P. Sridhar, S. Deb, and M. A. Islam. “DeEscalWild: A Real-World Benchmark for Automated De-Escalation Training with SLMs.” Manuscript submitted to the NeurIPS 2026 Evaluations and Datasets Track. First author; led benchmark construction, data validation, model fine-tuning, and evaluation.
5. **M. H. Hasan**, M. A. Jahan, M. E. Ali, Y.-F. Li, and T. Sellis. “A Multi-Modal Deep Learning Based Approach for House Price Prediction.” arXiv:2409.05335, 2024. First author; led multimodal architecture design and evaluation. [preprint]

## Industry Experience

### Machine Learning Engineer

*IQVIA, Dhaka, Bangladesh*

**May 2022 – Jul 2025**

*Full-time*

- Built and deployed ML prediction systems using Python, PyTorch, TensorFlow, and Snowflake; production work contributed to solutions associated with a 15% increase in sales.
- Developed time-series data pipelines over enterprise data marts, including data engineering, preprocessing, model optimization, evaluation, and deployment.
- Architected LangChain-based multi-agent workflows, fine-tuned open-source LLMs for domain-specific applications, and conducted AI security red-team assessments.
- Implemented scalable MLOps with AWS SageMaker/Bedrock, Kubernetes, Kubeflow, Docker, ECR/EC2, Git, and GitLab CI/CD.

## Teaching Experience

### Lecturer, Department of CSE

*Bangladesh University of Engineering and Technology (BUET)*

**Nov 2022 – Jul 2024**

*Part-time*

- Taught six undergraduate sessional/laboratory courses: Algorithm Engineering, Software Development, Databases, Data Structures and Algorithms II, Digital Logic Design, and Structured Programming.
- Supervised **30 student projects**; designed and graded assignments, laboratory exercises, and examinations for classes of 40–60 students.

## Research Grant

**Research Grant for M.Sc. Thesis**, Directorate of Advisory, Extension and Research Services, BUET, 2024. Competitive award of approximately **US\$1,500** supporting MapAgent research.

## Awards & Honors

- Platinum Award, IQVIA Impact Program, 2024; Silver Award, IQVIA Impact Program, 2025.
- University Merit List Scholarship (two semesters).
- Dean’s List for three consecutive years (Levels 2–4), BUET.

## Technical Skills

**Programming:** Python, C/C++, Java, JavaScript, SQL, LaTeX

**ML / AI:** PyTorch, TensorFlow, Hugging Face, scikit-learn, LangChain, LLM fine-tuning, RAG, agentic systems, model evaluation

**Cloud / Systems:** AWS SageMaker, Bedrock, EC2, ECR, Docker, Kubernetes, Kubeflow, Git/GitLab CI/CD, NVIDIA GPU/HPC workflows

**Databases / Web:** Snowflake, PostgreSQL, MySQL, MongoDB; Node.js, Vue.js, React

## References

**Mohammad Atiqul Islam**  
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**Mohammed Eunus Ali**  
Associate Professor, Monash University  
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